

Federated Learning & The Selection Problem

What is Federated Learning?

- Collaborative ML — data never leaves the client
- Clients train locally, share only model updates (gradients)
- Central server aggregates updates via FedAvg
- Non-IID data across clients
- Unbalanced data volumes
- Privacy preserved — raw data stays local

Problem: Full Participation is Infeasible

Challenges:

- Communication overhead at scale
- Slow / resource-constrained stragglers
- Biased updates hurt global convergence

Our Three Strategies:

- DivFL — Greedy Submodular diversity
- S-FedAvg — Shapley value contribution
- SW-ShUCB — Sliding Window Bandit

Client Selection Strategies For Stationary Data

Three algorithms evaluated on MNIST

DivFL

Greedy Submodular

Goal: Select subset S^* maximising diversity of client gradient updates.

$$F(S) = \sum_i \min_{j \in S} d(\nabla_i, \nabla_j)$$

Greedy addition:

Each step picks client most dissimilar to already-selected set.

Guarantee:

$(1 - 1/e)$ approximation to optimal via submodularity.

S-FedAvg

Shapley Values

Goal: Quantify each client's marginal contribution to model performance.

$$\phi_i = \sum [|S|! (n - |S| - 1)! / n!] [v(S \cup \{i\}) - v(S)]$$

Exact:

$O(2^n)$ — infeasible for large n .

Solution:

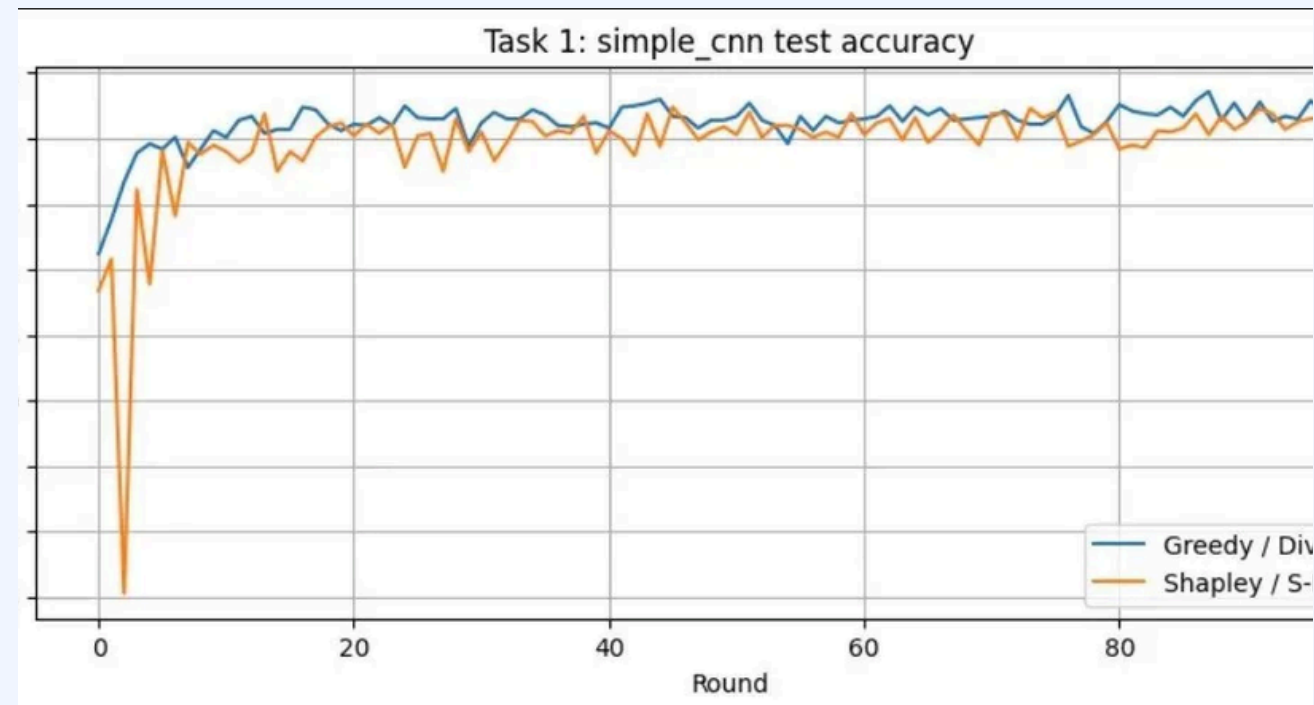
Monte Carlo sampling of permutations to estimate ϕ_i .

Selection:

High- ϕ_i clients selected; negative- ϕ_i excluded as noisy.

Task 1: Baseline Results (No Drift)

DivFL vs S-FedAvg · 5 models 100 clients · K=10 · MNIST · 100 rounds



Model	DivFL	S-FedAvg
Simple CNN	99.22%	99.10%
LeNet	98.98%	98.86%
Random Forest	93.98%	92.50%
SVM	91.33%	91.13%
Logistic Reg.	88.70%	86.86%

✓ Key Insights

- DivFL consistently leads across all models
- CNNs achieve >99% accuracy
- Both strategies converge well without drift

Concept Drift in Federated Learning

Why static selection fails in non-stationary environments

Concept Drift: The data distribution $P(X,Y)$ changes over time, violating the stationarity assumption of standard FL. Client updates that were once informative may become stale, irrelevant, or harmful to the global model.

Label Distribution Shift

Class proportions change across rounds.

Ex: Balanced digits 0–9 → predominantly 1 & 7.

Effect:

Model loses accuracy on unseen classes. DivFL & S-FedAvg cannot detect nor recover — CNN accuracy collapses to <6%.

Feature Distribution Shift

Input space $P(X)$ changes while labels stay consistent.

Ex: Pixel brightness dims gradually; Gaussian noise injected.

Effect:

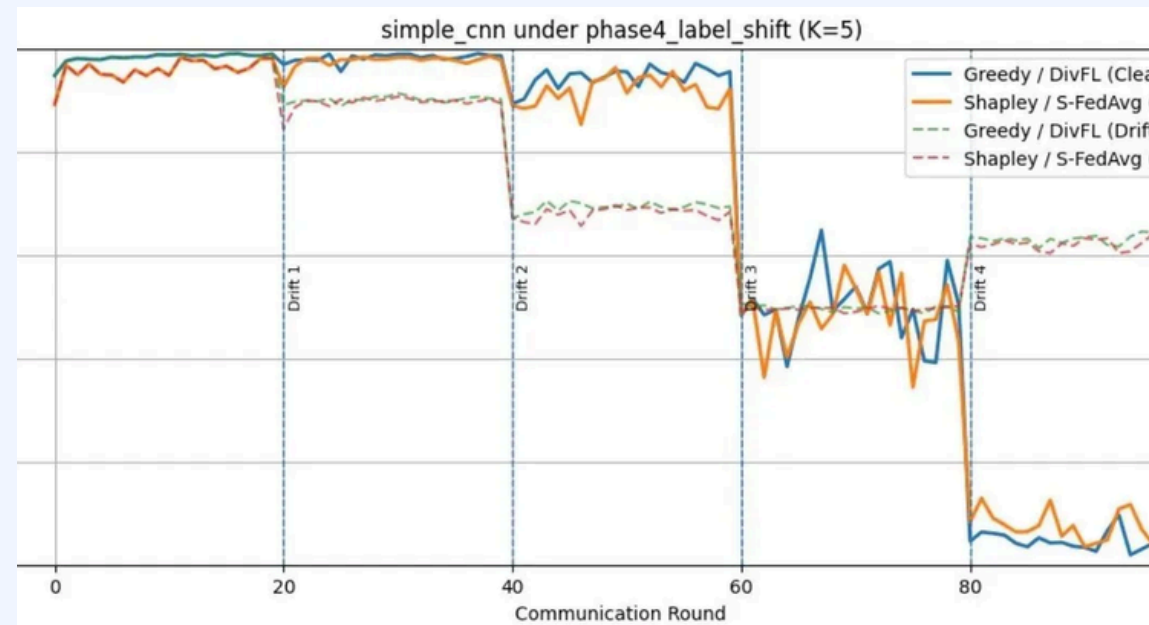
Model degrades more slowly — CNN retains ~90% accuracy since structure generalises to shifted inputs.

Both DivFL & S-FedAvg assume stationary reward signals — Shapley values computed on pre-drift data are stale; greedy diversity ignores temporal relevance. Neither can detect or recover → motivates SW-ShUCB.

Task 2: Performance Under Concept Drift

Phase-4 drift events at rounds 20, 40, 60, 80

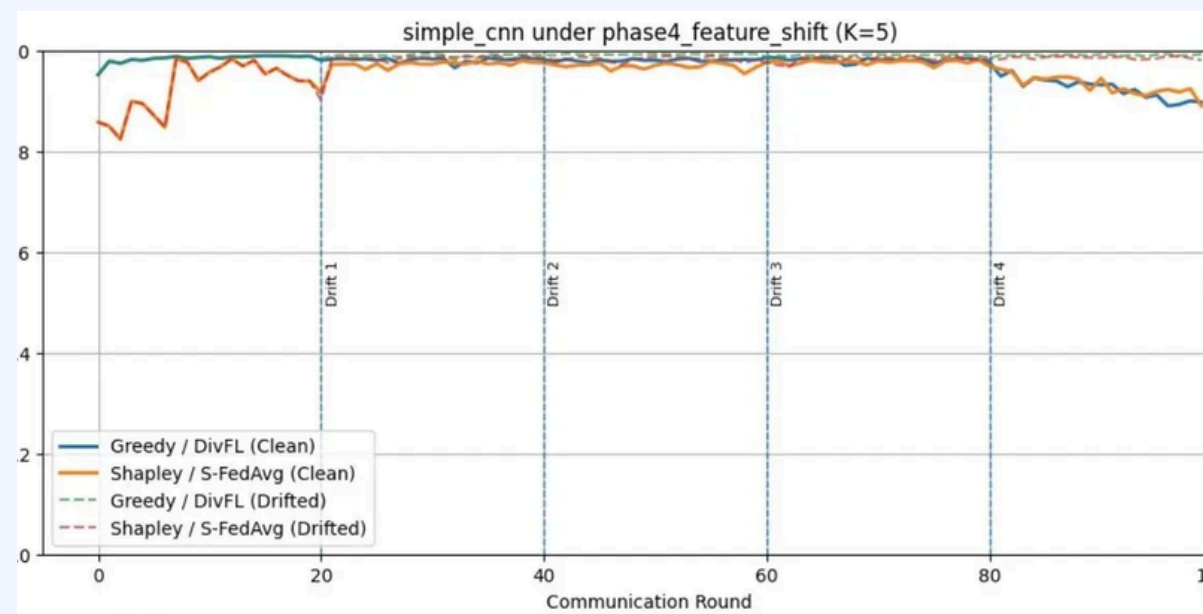
Label Shift – Final Accuracy (%)



Label drift is catastrophic

Accuracy collapses to near-random levels because label permutations break the learned class mapping.

Feature Shift – Final Accuracy (%)



Feature drift is manageable

Accuracy stays around 90% because the class mapping is unchanged and only the input distribution shifts.

Both methods lack drift-awareness

Neither DivFL nor S-FedAvg can detect or recover
→ motivates SW-ShUCB.

Novelty: From MAB to SW-ShUCB

Addressing non-stationary client selection with a sliding-window bandit strategy

Multi-Armed Bandit (MAB)

Each client = one arm. Server selects K clients; receives reward reflecting update quality. Gradually learns best clients.

Limitation: Assumes stationary rewards — once a client's data drifts, old reward estimates mislead selection.

✗ Stationary rewards only

UCB — Upper Confidence Bound

Balances exploration vs exploitation:

$$\text{Score}(i) = \bar{x}_i + \sqrt{(2 \ln(t) / n_i)}$$

High-uncertainty clients explored; well-known clients exploited.

Limitation: n_i accumulates across ALL rounds — old stale observations are never forgotten after drift.

✗ Stale counts — no forgetting

SW-ShUCB (Our Method)

Two key innovations:

1. Sliding Window W — only last W rounds retained; old rewards discarded. Naturally adapts to drift.
2. Shapley reward ϕ_i — marginal contribution as reward signal, capturing true utility.

$$\text{Score}(i) = \phi_i^-(W) + \sqrt{(2 \ln(t) / n_i(W))}$$

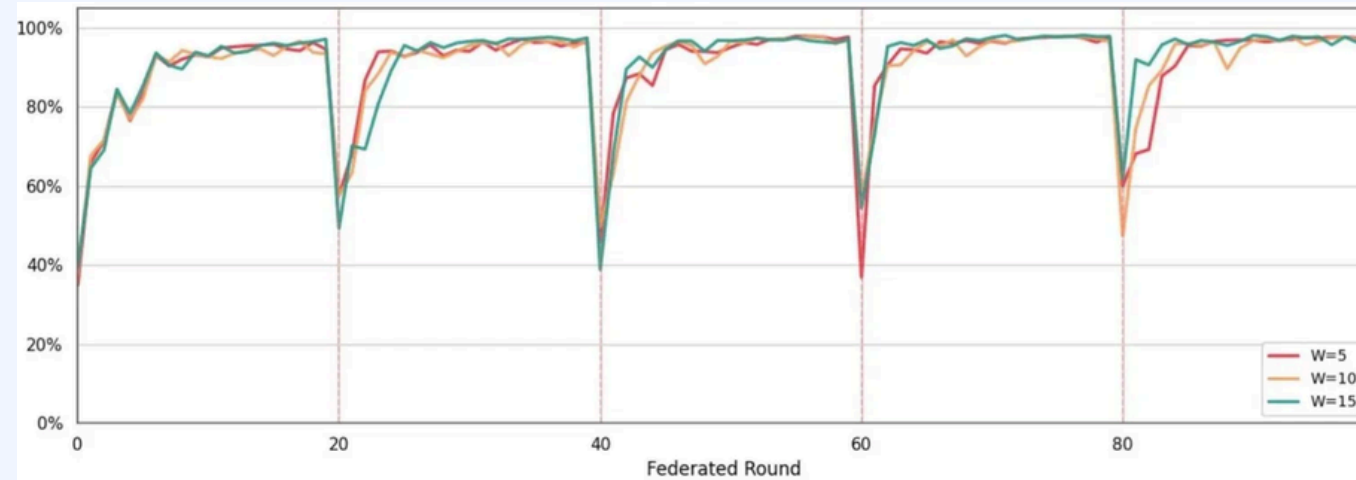
Top-K clients by Score selected each round.

✓ Non-stationary aware

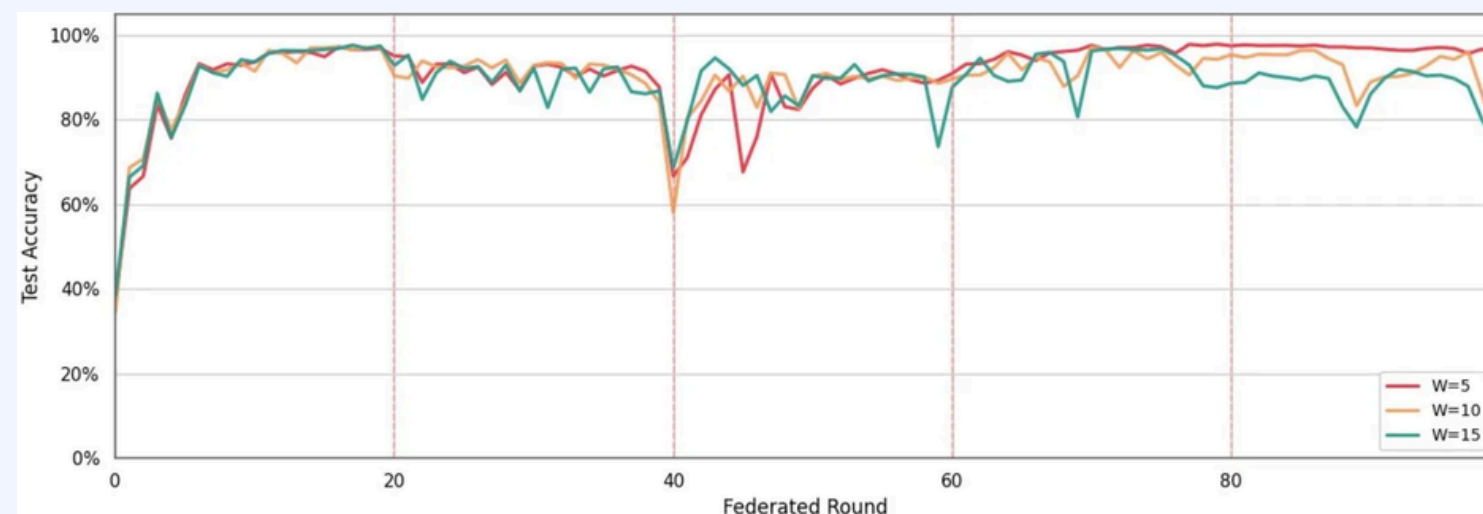
Task 3: SW-ShUCB Results

Simple CNN · $K=10$ · $W \in \{5, 10, 15\}$ · 100 clients · 100 rounds · MNIST

Label Drift — Accuracy by Window Size



Feature Drift — Accuracy by Window Size



SW-ShUCB Analysis

Drift-aware client selection keeps recent rewards relevant

Rapid recovery after drift

Accuracy rebounds quickly after each change point, staying around 97–98%. This shows the sliding window keeps the algorithm responsive

Main takeaway

Static client selection becomes stale under non-stationarity. SW-ShUCB uses recent Shapley rewards plus UCB exploration to learn performance trends.

Window-size trade-off

W = 15 is slightly more stable at the peak; W = 5 adapts faster but is a bit noisier after drift.

Best balance

W = 10 or W = 15 usually gives the best overall performance.

★ SW-ShUCB recovers quickly after drift

97%+ accuracy vs <6% → ~16× improvement, with zero extra communication overhead.

★ Window-size trade-off

Label drift: larger W (15) is slightly more stable.
Feature drift: smaller W (5) adapts faster after change points.

Conclusion & Future Directions

Task 1

Baseline (No Drift)

DivFL edges S-FedAvg across all 5 models. Simple CNN achieves 99.22% — both strategies converge well under stationary IID-like conditions.

Task 2

Under Concept Drift

Both strategies fail catastrophically under label drift (<6% CNN). Feature drift more tolerable (~90%). Static selection is insufficient for non-stationary FL.

Task 3

SW-ShUCB Recovery

97%+ accuracy under label drift — ~16× improvement. Sliding-window UCB + Shapley rewards enables drift-aware selection at zero extra bandwidth cost.

Core Insight: Drift-aware client selection is essential for robust real-world Federated Learning.

→ **Adaptive Window:** Dynamically tune W using ADWIN / Page-Hinkley drift detection — avoids the W trade-off entirely.

→ **Personalised FL:** Extend SW-ShUCB to per-client local adaptation (pFedMe) for heterogeneous drift rates.

→ **Convergence Theory:** Derive formal convergence bounds for SW-ShUCB under non-stationary distributions.

→ **Real-World Benchmarks:** Validate on CIFAR-10, FEMNIST, and real IoT/sensor streams with organic concept drift.